higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE (VOCATIONAL)

ENGINEERING FABRICATION – SHEET METAL WORK
NQF LEVEL 3

(6031023)

11 March 2019 (X-Paper)
09:00–12:00

This question paper consists of 10 pages and 1 answer sheet.

NCV

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TIME: 3 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. QUESTION 1.10 must be answered on the ANSWER SHEET (attached).
5. Write neatly and legibly.

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QUESTION 1

1.1 State ONE example of each of the following metal categories:

1.1.1 Ferrous metals

1.1.2 Nonferrous metals

1.1.3 Alloys

(3 × 1) (3)

1.2 Choose a physical or chemical property from COLUMN B that matches a metal in COLUMN A. Write only the letter (A–E) next to the question number (1.2.1–1.2.5) in the ANSWER BOOK.

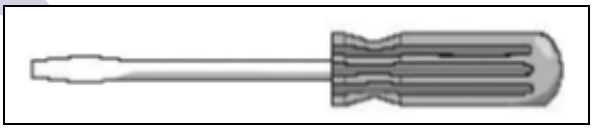
COLUMN A METALS		COLUMN B PROPERTIES	
1.2.1	Copper	A	good magnetic properties
1.2.2	Steel	B	good conductor of heat
1.2.3	Brass	C	very brittle
1.2.4	White cast iron	D	very ductile
1.2.5	Solder	E	low melting point

(5 × 1) (5)

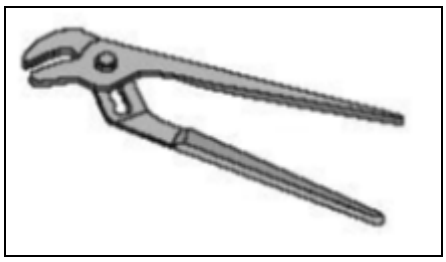
1.3 The following sketches indicate certain hand tools and machines used during sheet-metal work processes.

Name each tool and state ONE use of each.

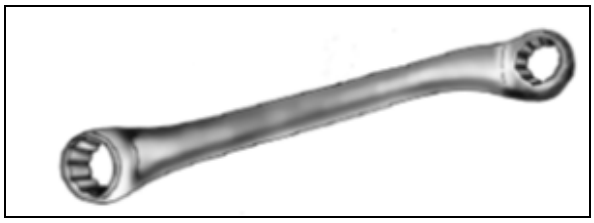
1.3.1



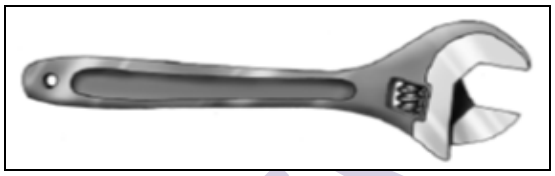
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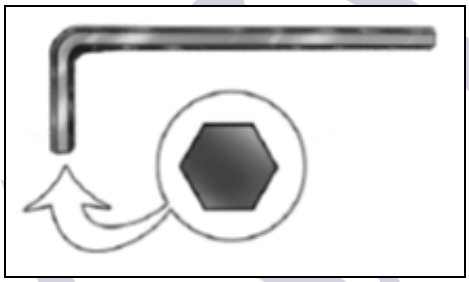
1.3.3



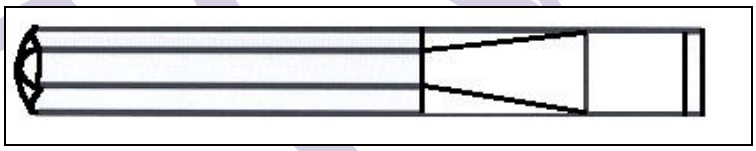
1.3.4



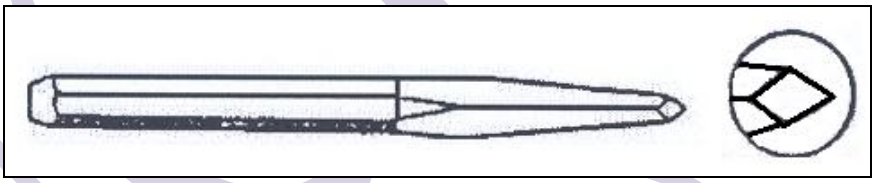
1.3.5



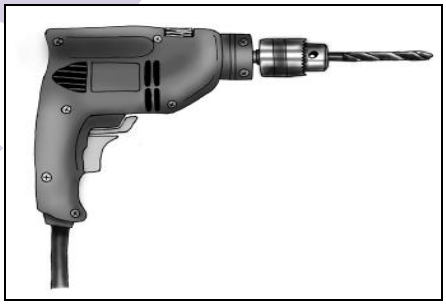
1.3.6



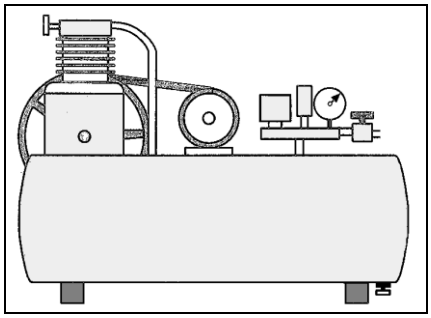
1.3.7



1.3.8



1.3.9

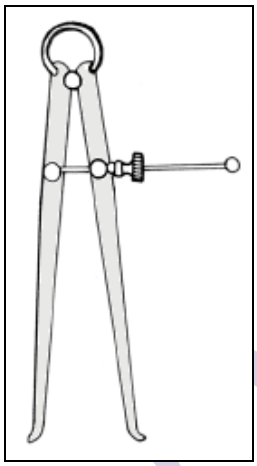


(9 × 2) (18)

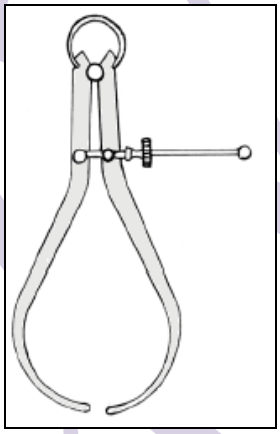
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1.4 Name the following measuring tools and state ONE use of each:

1.4.1



1.4.2



(2 × 2) (4)

1.5 In FIGURE 1 a combination set is shown. Each component can be used in conjunction with the ruler to accomplish certain tasks.

Name each component indicated, by writing only the name of the component next to the question number (1.5.1–1.5.3) in the ANSWER BOOK.

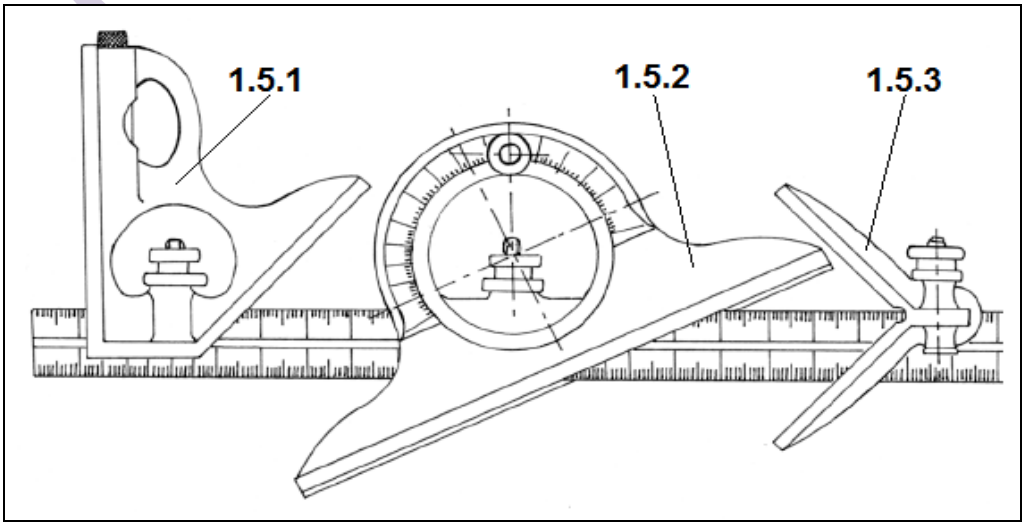


FIGURE 1 (3)

1.6 FIGURE 2 and FIGURE 3 show two of the components of the combination set.

Name each component and state the process that each one is used for.

1.6.1

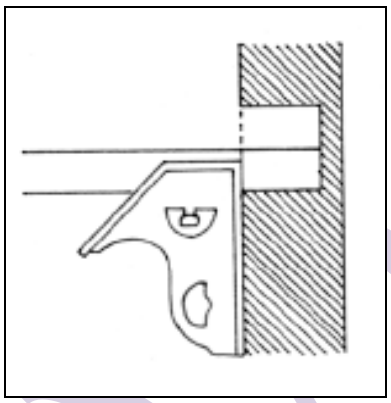


FIGURE 2

1.6.2

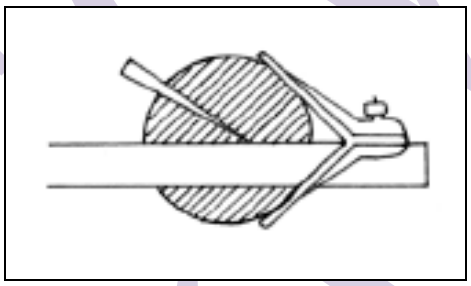


FIGURE 3

(2 × 2) (4)

1.7 Make neat drawings illustrating the difference between the included angles of the *prick punch* and the *centre punch*. Name each punch and the angle sizes.

(2 × 2) (4)

1.8 Draw a line piece of ± 50 mm long. Do the necessary construction to bisect the line piece. (3 × 1) (3)

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1.9 FIGURE 4 shows the development of a round pipe.

Give the formula to determine length A.

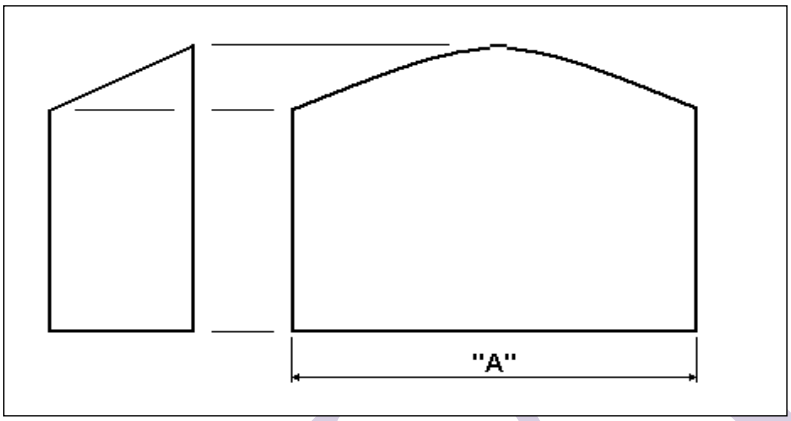


FIGURE 4 (1)

1.10 The front view of a round oblique pipe connection is shown on the ANSWER SHEET (attached). Make use of this drawing and construct the development of the round oblique pipe. Your drawing only has to adhere to correctness of the development and good proportion. Make use of the reference line as indicated.

NOTE: Write your EXAMINATION NUMBER in the space provided on the ANSWER SHEET, detach it from the question paper and hand it in with your ANSWER BOOK. (6 × 1) (6)

1.11 FIGURE 5 indicates the floor plan of a workshop.

Use Pythagoras' theorem to calculate the total area of the floor, by using the diagonal length.

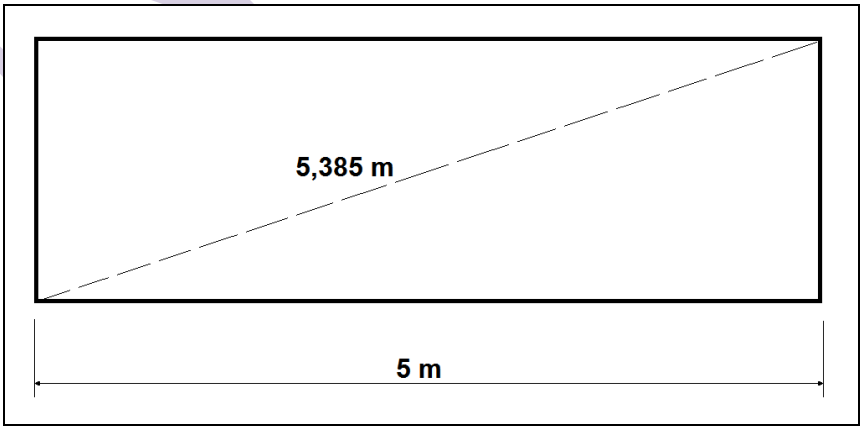


FIGURE 5 (6 × 1) (6)

1.12 Describe the term *isometric drawing* applicable to drawing projections. (2)

1.13 FIGURE 6 indicate a specific notch used during sheet-metal work. Name the specific type of notch.

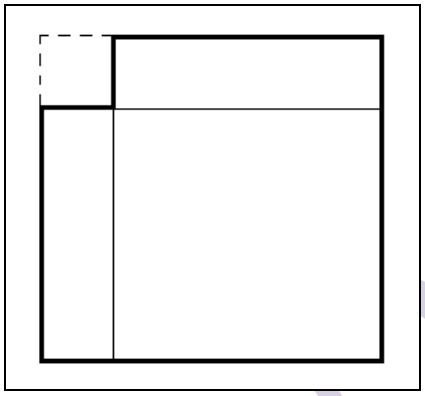


FIGURE 6 (1)

[60]

QUESTION 2

2.1 State FOUR safety precautions with regard to flame cutting work. (4 × 1) (4)

2.2 FIGURE 7 and FIGURE 8 indicate oxyacetylene pipes with connectors.

Identify which connector will fit on the *oxygen system* and which connector will fit on the *acetylene system*. Write the type of gas system next to the correlating figure number (FIGURE 7 and FIGURE 8) in the ANSWER BOOK.

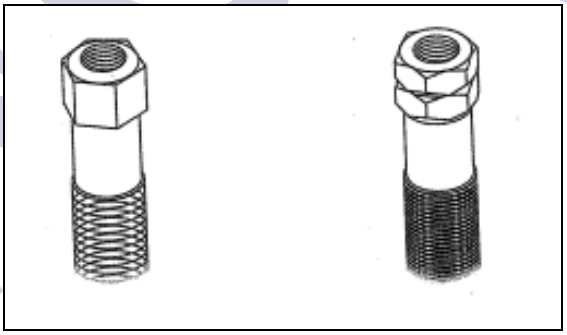


FIGURE 7 **FIGURE 8** (2 × 1) (2)

2.3 State the TWO main factors that influence the effective shape of the flame and the heat values during gas welding with regard to the nozzle. (2 × 1) (2)

2.4 Describe the following terms related to gas welding:

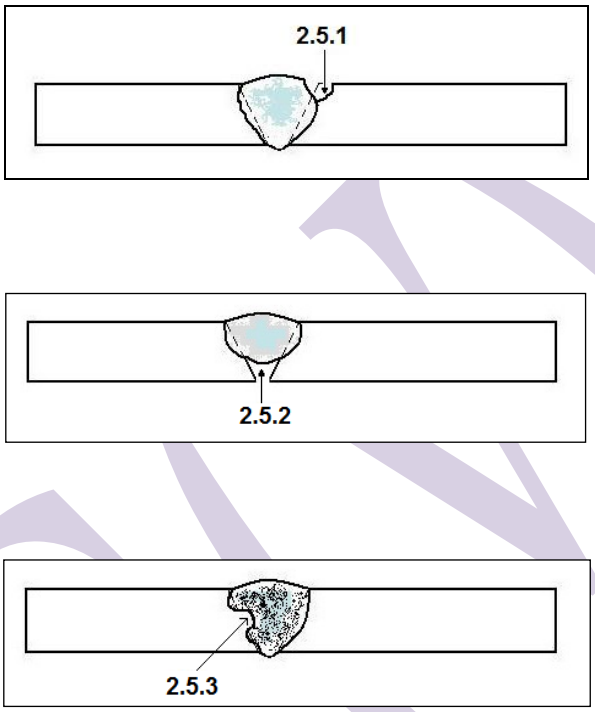
2.4.1 Backfiring

2.4.2 Flashback

(2 × 2) (4)

2.5 Different gas-welding defects can occur during the gas-welding process due to many reasons, for example scale on the surface of the workpiece.

Name the defect illustrated in each of the following drawings by writing only the answer next to the question number (2.5.1–2.5.3) in the ANSWER BOOK.



(3 × 1) (3)

2.6 During and after using welding equipment and tools, damages can occur unexpectedly.

State FIVE caring and storing procedures that must be followed to ensure that damage DOES NOT occur to welding equipment and tools. (5 × 1) (5)

[20]

QUESTION 3

3.1 Explain the meaning of scale 1:100 applicable to drawing convention. (2)

3.2 It is very important to plan the correct type of ventilation system due to pollutants that have a big influence on the pureness of conditioned air.

State FOUR pollutants that can cause problems in an HVAC system. (4 × 1) (4)

3.3 During planning and preparation to perform a task, specific information is needed.

State THREE information standards that must be known before a task can commence. (3 × 1) (3)

3.4 It is crucial to have a checklist to ensure that components are correctly marked and it is essential once preparations must be done.

State THREE points that will be on a checklist. (3 × 1) (3)

3.5 Crane drivers are responsible for lifting and moving objects safely therefore they have to be very careful. To adhere to standards, certain rules apply for safe working procedures.

State FOUR safe working procedures or rules that crane drivers have to adhere to. (4)

3.6 Fibre such as polypropylene is used to support the core of lifting ropes.

Give FOUR reasons why the inner core must be supported. (4 × 1) (4)

[20]

TOTAL: 100

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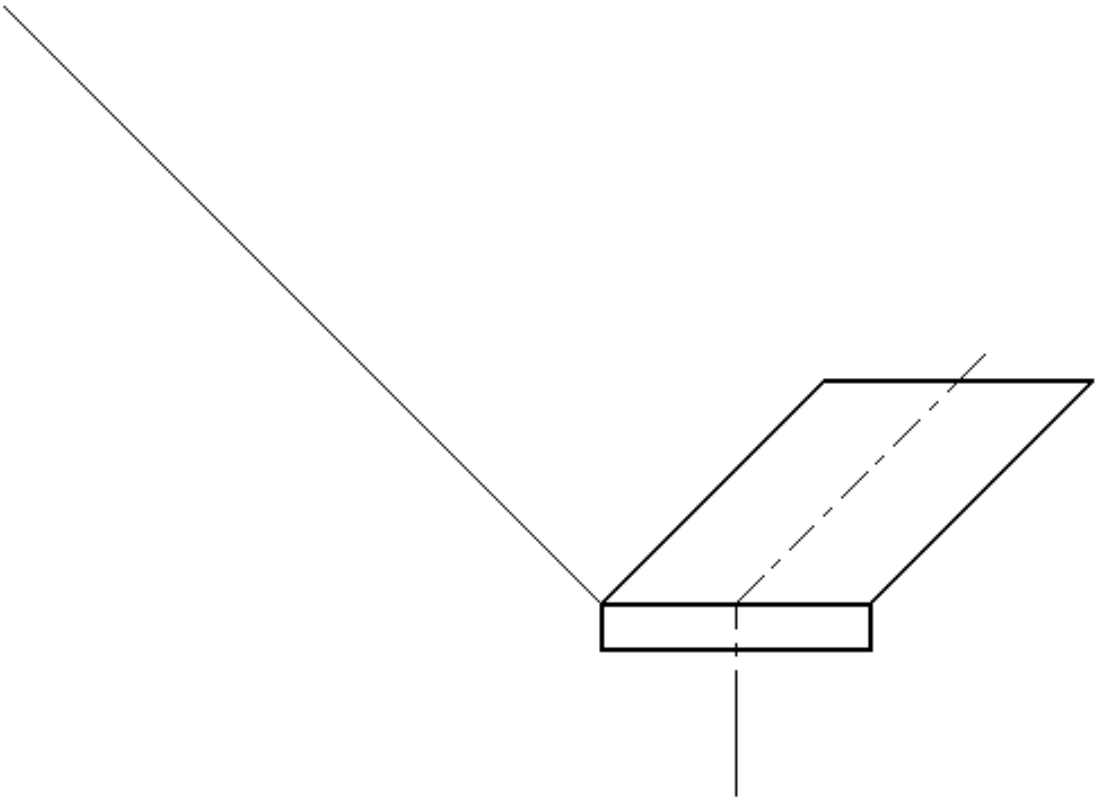
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ANSWER SHEET

EXAMINATION NUMBER: | | | | | | | | | | | | | |

CENTRE NUMBER: | | | | | | | | | | |

QUESTION 1.10



NOTE: Write your EXAMINATION NUMBER in the space provided on the ANSWER SHEET, detach it from the question paper and hand it in with your ANSWER BOOK.

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